Report

Of

AI Automation Internship

Lead Generation and Outreach System

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of B.Tech CSE

By:

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Batch: 2024-28

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CANDIDATE'S DECLARATION

I, Abhay Singh Nagarkoti, hereby declare that the internship report titled "AI Automation Internship: Lead Generation and Outreach System" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The work presented in this report is based on the internship completed at CashCow Labs between 8 June 2026 and 31 August 2026. All references used in preparing this report have been appropriately acknowledged. I further declare that this report has not been submitted to any other institute for the award of any degree or diploma, and I shall be solely responsible for any copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am thankful to CashCow Labs for giving me the chance to work on a live automation system rather than a training exercise. Being handed a real problem, with real data and real constraints, taught me more in twelve weeks than I expected.

I am especially grateful to Sharique, CEO and Co-Founder of CashCow Labs, for the guidance and the honest feedback through the internship period. Several parts of the system were rebuilt because of that feedback, and they came out better for it.

I also thank the School of Computer Science and Engineering, IILM University, Greater Noida, for the academic support and for encouraging students to take up industry internships alongside coursework.

Finally, I thank my parents and my friends for their support during the internship and while preparing this report.

Signature

Student Name: ABHAY SINGH NAGARKOTI

Roll No: 2410030450

INTERNSHIP COMPLETION CERTIFICATE

CashCow Labs.

CERTIFICATE

OF COMPLETION

PROUDLY PRESENTED TO

ABHAY SINGH NAGARKOTI

Congratulations on successfully completing a 12-week internship as an **AI Automation Intern**, from June 8th, 2026, to August 31st, 2026. Your attention to detail has been amazing.

Sharique

CEO/CO-FOUNDER

Issued by CashCow Labs for a 12 week internship as AI Automation Intern, from 8 June 2026 to 31 August 2026.

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1. INTRODUCTION

An internship is usually the first time a student works on a problem that does not come with a fixed answer. A college assignment has a known input and an expected output. A business system does not. It runs on data that is incomplete, it depends on services that go down without warning, and it has to keep working when nobody is watching it. Getting used to that gap is the main reason internships matter.

This report covers a twelve week internship completed at CashCow Labs as an AI Automation Intern, from 8 June 2026 to 31 August 2026. The work was in workflow automation and applied AI. The platform used for orchestration was n8n, a low code automation tool that connects APIs, databases and language models into a single running workflow.

The project built during the internship is a Lead Generation and Outreach System. In plain terms, it does the job of a junior sales researcher. It finds people on LinkedIn who match a customer profile the company defines, checks whether each person is actually relevant, reads up on their company and their competitors, works out what problems that company is likely to have, writes a cold email that refers to those specific problems, sends it, and follows up three days later if there is no reply.

The finished workflow contains 179 nodes and runs on seven independent triggers. It is not a demo. It was built to run unattended on a schedule and to recover on its own when a step fails.

Chapters 2 and 3 cover the host organization and how the twelve weeks were spent. Chapters 4 to 6 set out the problem the system solves and what was and was not included. Chapters 7 to 10 are the technical core: the tools used, how the system is put together, the approach that was followed, and what was actually built. Chapters 11 to 14 cover what was learned, what went wrong along the way, and where the system could go next.

2. ORGANIZATION PROFILE

CashCow Labs is a limited liability company registered in the state of Wyoming, United States. The company works with B2B, D2C and B2C businesses and helps them grow by automating the processes that normally eat up the most manual hours. Prospecting, outreach, content operations and reporting are the usual targets.

The approach the company takes is worth describing, because it shaped how the project was built. The company does not start by picking a tool. It starts by mapping how a task would be done by a careful human being, step by step, and only then decides which of those steps can be handed to software and which ones need a person. Automation that skips this step tends to produce output that is fast and useless.

Organization Detail	Information
Organization	CashCow Labs
Legal Structure	Limited Liability Company (LLC)
Registered In	Wyoming, United States
Domain	AI Automation and Business Process Automation
Clients Served	B2B, D2C and B2C businesses
Mode of Internship	Remote
Duration	12 Weeks (8 June 2026 to 31 August 2026)
Role	AI Automation Intern
Reporting To	Sharique, CEO and Co-Founder

CashCow Labs has delivered work for enterprise software firms, fintech companies, media businesses and independent creators. The range is wide on purpose, because the underlying problem in each case is the same one: a repeatable process that is being done by hand at a scale where hands no longer work. A short note on each is given below.

Client	Sector	Brief
Sirion	Enterprise Software	Contract lifecycle management platform founded in 2012. Its AI native system is used to author, negotiate and govern enterprise contracts, and the company reports managing over 7 million contracts across more than 100 languages.
Wise	Fintech	Cross border payments and multi currency banking, founded in London in 2011. Wise reported around 19 million active customers and 243.5 billion dollars in cross border volume for the financial year ending March 2026, and is listed on both the London Stock Exchange and Nasdaq.
Ali Abdal	Creator Economy	Creator, author and podcaster in the productivity space. A Cambridge trained doctor who moved into content full time, with a YouTube channel of roughly 6.7 million subscribers and a bestselling book, Feel-Good Productivity.
Greg Isenberg	Venture and Community	Entrepreneur and angel investor, CEO of the holding company Late Checkout. He has founded and exited three community focused companies and previously advised Reddit and TikTok.
BenAI	AI Automation	AI automation agency run by Ben van Sprundel, who also publishes build tutorials on the Ben AI YouTube channel covering n8n, AI agents and no-code automation systems.
Nas Daily	Media	Media company founded in 2016 by Nuseir Yassin around one minute daily videos. It has grown into a group that includes Nas Academy and Nas.io, with roughly 14 million YouTube subscribers and over 60 million followers across platforms.

Client	Sector	Brief
SlidesAI.io	AI Productivity	An AI presentation tool that turns text, notes and prompts into slide decks, with native integrations for Google Slides and PowerPoint and over 14 million reported installs.

Working inside an organization with this client mix meant the system being built could not be tuned to one industry. A pipeline that only works for SaaS founders is not much use to a company whose next client might be a media group. This constraint pushed the design towards keeping the customer profile as configurable data rather than as logic written into the workflow, which is covered in Chapter 9.

3. INTERNSHIP OVERVIEW

- Internship Mode: Remote
- Duration: 12 Weeks, 8 June 2026 to 31 August 2026
- Domain: AI Automation and Business Process Automation
- Role: AI Automation Intern
- Organization: CashCow Labs, Wyoming, United States
- Reporting To: Sharique, CEO and Co-Founder
- Project Title: Lead Generation and Outreach System
- Project Focus: Automated prospect sourcing, research, personalised email drafting and follow up.

The internship ran for twelve weeks in remote mode. The first two weeks were spent learning the platform and the company's way of working. From week three onward the work was project based, with each stage of the Lead Generation and Outreach System built, tested and reviewed before the next stage was started.

Building the system stage by stage rather than all at once turned out to be important. Each stage writes its result into Airtable and sets a flag. The next stage picks up only the records whose flag is set. This meant a broken stage never took the whole system down with it, and it also meant a stage could be reviewed and rebuilt in isolation.

Week	Focus	Outcome
1 to 2	Onboarding, n8n fundamentals, Airtable schema design	Base structure created with five linked tables and the fields each stage would need to read and write
3 to 4	Lead sourcing	Three sourcing routes built and made switchable: Apify LinkedIn actors, Google search operators, and Prospeo people search

Week	Focus	Outcome
5 to 6	Lead qualification	Language model agents with structured output parsers that return a qualified or disqualified verdict with a written reason
7 to 8	Company and competitor research	Company detail scraping and a research layer built on Jina Search and Jina Reader
9	Profile intelligence	Chain that produces a written analysis, pain points, matched solutions and engagement hooks for each lead
10	Email enrichment and drafting	Prospeo enrichment for verified email addresses, and an agent that produces three distinct drafts per lead
11	Sending and follow up	Gmail sending on a daily schedule, reply detection, and a follow up agent that fires after three days of silence
12	Testing, error handling and handover	Wait nodes for rate limits, retry paths, duplicate protection, and documentation of the full workflow

Work was reviewed against the output of the system, not against the number of nodes built. A stage was accepted when the records it produced in Airtable were correct and readable by a person, and rejected when they were not. This is a harder standard than it sounds, because a workflow can execute without error and still produce output that is wrong.

4. PROBLEM STATEMENT

Outbound sales is repetitive, but it is not simple. Doing it properly for a single prospect involves a sequence of small judgements. Find people who match the profile the business is targeting. Check that the profile is current, because a person's headline often describes a role they left two years ago. Look up the company and understand what it sells and who it sells to. Work out which problems that company is likely to be facing. Write an email that refers to those specific problems rather than to a generic pitch. Send it. Remember to follow up if nothing comes back, and remember not to follow up with someone who already replied.

Done carefully, that is fifteen to twenty minutes per prospect. At that rate a person manages perhaps twenty five prospects in a working day, and the quality of the research drops as the day goes on. The usual response to this is to give up on research and send the same email to a thousand people. That approach is cheap and it is why most cold email gets ignored.

The gap sits between two bad options. Manual research produces good emails at a volume too low to matter. Mass mailing produces volume with no relevance. Existing sales tools mostly automate the sending, which is the easiest part, and leave the research to the human.

The problem taken up during this internship was to build a system that keeps the quality of manual research while removing the manual time, and that can be pointed at a completely new customer profile without any of its logic being rewritten. It also had to run unattended and recover from failure on its own, because a system that needs a person to restart it after every API error is not really automated.

5. PROJECT OBJECTIVES

Seven objectives were set at the start of the project. Each one maps to a stage of the finished system.

1. Source leads from more than one channel. Depending on a single provider means the whole pipeline stops when that provider blocks, rate limits or changes its output format.
2. Qualify every lead automatically against a customer profile stored as data, so that the profile can be changed without touching the workflow.
3. Research each prospect and their company, including their competitors, so that the email has something specific to refer to.
4. Find a verified email address for each qualified lead, and skip the lead rather than guess when no verified address exists.
5. Generate personalised drafts rather than filling a template, and generate more than one so that a choice can be made.
6. Send on a schedule and follow up after a fixed period of silence, without ever following up with someone who has already replied.
7. Keep every intermediate result in one database so that any lead can be traced from the moment it was found to the moment it was contacted.

6. SCOPE OF THE PROJECT

The project is scoped to automated outbound prospecting for a defined customer profile. The following work is included.

- Lead sourcing from LinkedIn through Apify actors, from search engines through structured search operators, and from Prospeo's people search index.
- Automatic qualification of each lead against a stored Ideal Customer Profile, with a written reason recorded for both acceptance and rejection.
- Company research covering description, industry, size, location, specialities, target market and key services.
- Competitor research, including keyword generation, competitor discovery and a written competitive summary.

- Generation of a profile analysis, pain points, matched solutions and engagement hooks for each qualified lead.
- Email address enrichment restricted to verified addresses only.
- Generation of three email drafts per lead, selection of one, and sending through Gmail on a daily schedule.
- Reply detection and a single follow up email after three days of no response.
- Full state tracking in Airtable across five linked tables.

The following areas were deliberately left outside the current scope and are noted in Chapter 14 as possible extensions.

- Reply handling and conversation management after the first response. The system detects that a reply exists and stops. It does not attempt to answer.
- Meeting booking and calendar integration.
- Multi channel outreach. The system uses email only. LinkedIn messaging and phone were not included.
- A user interface. The system is operated through Airtable and the n8n editor.
- Deliverability infrastructure such as domain warm up, inbox rotation and bounce handling. These were treated as an operational concern outside the workflow.

7. TECHNOLOGIES AND TOOLS USED

The system uses nine external services. Each was chosen for a specific job, and in most cases because the obvious alternative had a problem that mattered.

Layer	**Technology**	**Role in the System**
Orchestration	n8n	Runs the entire workflow. Handles triggers, branching, looping, batching, waiting and error paths. Self hosted.
Lead sourcing	Apify	Four actors are used: LinkedIn Profile Search, LinkedIn Profile Scraper, LinkedIn Company Details, and a low cost Google search results scraper. The LinkedIn actors used are cookie free, which removes the need to hold a session token.
Contact data	Prospeo API	Two endpoints. The search-person endpoint returns people matching job title, headcount, industry and location filters. The enrich-person endpoint returns a verified email address for a given name and company domain.
Web research	Jina AI	Two endpoints. s.jina.ai performs a search and returns readable results. r.jina.ai fetches a page and returns clean text instead of raw HTML, which matters because the output is being fed to a language model.

Layer	Technology	Role in the System
Language models	OpenAI GPT-4.1-mini and GPT-4o-mini	Fifteen model nodes across the workflow. GPT-4.1-mini handles reasoning heavy stages such as qualification and analysis. GPT-4o-mini handles lighter formatting work.
Agent framework	n8n LangChain nodes	Eleven agent nodes and four LLM chain nodes. Agents are given tools, such as the Jina endpoints and the Gmail sender, and decide when to call them.
Output validation	Structured Output Parsers	Four parsers enforce a JSON shape on model output. Two are set to auto fix, which sends malformed output back to the model with the schema attached rather than failing the run.
Database	Airtable	Forty node operations across five tables. Airtable holds the customer profile, every lead, all research output, all drafts and all status flags.
Email	Gmail	Sending is done through Gmail tool nodes attached to agents. A separate Gmail node performs the reply search before follow ups are drafted.

The workflow contains 179 nodes in total, of which 171 are functional and 8 are sticky notes used for documentation inside the editor. The breakdown below gives a sense of where the weight sits.

Node Type	Count	Purpose
Airtable	40	Read, create, update and upsert operations across five tables
OpenAI Chat Model	15	Language model nodes attached to agents and chains
No Operation	15	Terminating branches, used to make the false path of a condition explicit
Set (Edit Fields)	13	Field mapping and reshaping between stages
AI Agent	11	Qualification, research, analysis, drafting, sending and follow up agents
Code (JavaScript)	11	Date maths, JSON restructuring and output cleanup
Split In Batches	10	Loop control so that large record sets are processed a few at a time
If	9	Conditional branching, mostly guarding against empty or duplicate records
Wait	7	Deliberate pauses inserted to stay inside third party rate limits
Schedule Trigger	6	Independent schedules for six of the seven pipelines
HTTP Request Tool	6	Jina endpoints exposed to agents as callable tools
Structured Output Parser	4	JSON schema enforcement on model output
LLM Chain	4	Single shot generation without tool access
Apify	4	The four scraping actors

Node Type	Count	Purpose
Others	24	Merge, Switch, Split Out, Gmail, HTTP Request, Webhook, Remove Duplicates, triggers and sticky notes

8. SYSTEM ARCHITECTURE

The system is not one long chain. It is seven independent pipelines that share a database. Each pipeline has its own trigger and its own schedule, reads the records that are ready for it, does its work, writes the result back and sets a flag. No pipeline calls another pipeline directly.

This is the single most important decision in the design and the reason for it is practical. A lead passes through scraping, qualification, company research, competitor research, enrichment, analysis, drafting, sending and follow up. Run as one chain, that is a workflow lasting several minutes per lead that fails completely if any one API call times out. Split into seven, a failure in the enrichment stage leaves every record that already passed qualification safely stored, and the enrichment stage simply picks them up on its next run.

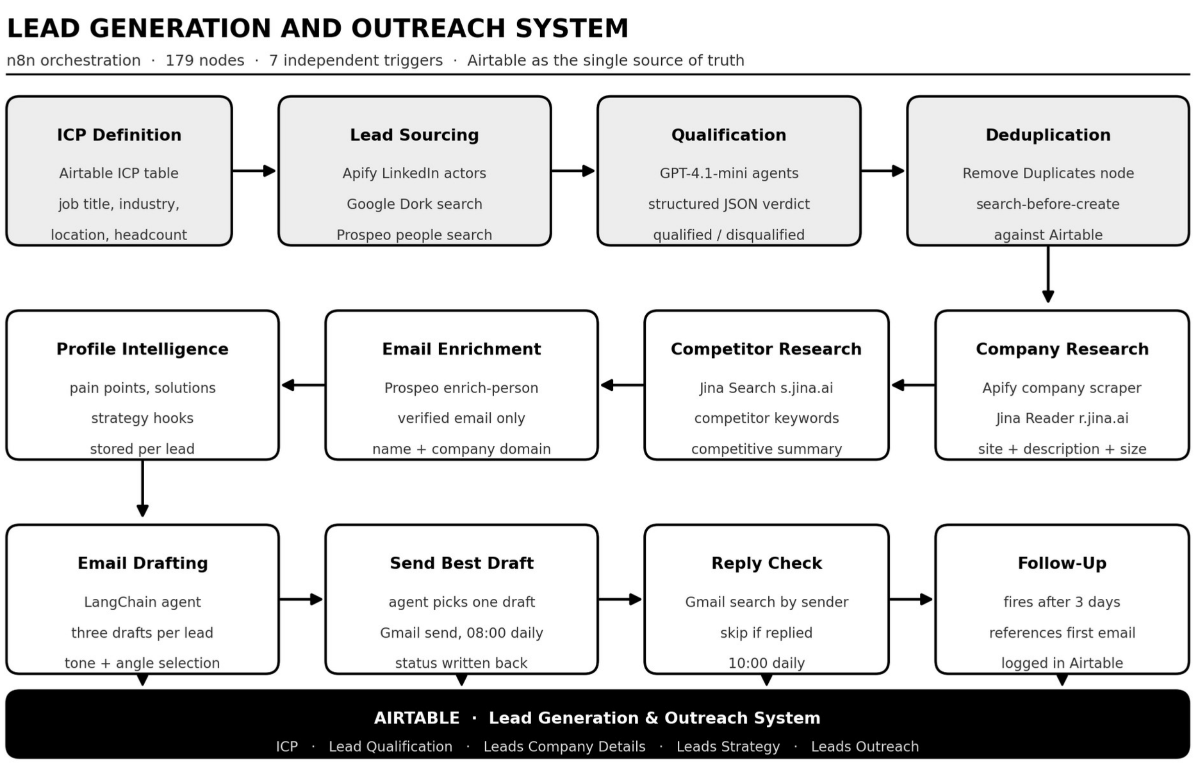

Figure 8.1: End to end architecture of the Lead Generation and Outreach System

8.1 The Seven Pipelines

Pipeline	Trigger	Nodes	What it does
1. Lead Sourcing and Qualification	Webhook	46	Routes each profile to one of three sourcing methods, scrapes, deduplicates and qualifies
2. Company and Competitor Analysis	Every 3 hours	32	Scrapes company details, researches the site and competitors through Jina
3. Profile Intelligence	Hourly	18	Writes an intelligence document, then derives pain points, solutions and hooks
4. Email Enrichment	Every 6 hours	13	Extracts the domain and calls Prospeo for a verified address
5. Email Drafting	Every 3 hours, plus manual	11	Feeds the research package to a drafting agent and stores three drafts
6. Email Sending	Daily at 08:00	7	Selects the strongest draft, sends it and records the sent date
7. Follow Up	Daily at 10:00	9	Checks Gmail for a reply after three days and follows up only where there is none

8.2 Data Layer

Airtable holds everything. The base is called Lead Generation and Outreach System and contains five tables.

Table	Holds
ICP	The Ideal Customer Profile definitions. Job title, decision maker, location, company size, industry, sourcing method and a flag for whether leads should currently be generated for that profile.
Lead Qualification	Every scraped lead with name, headline, about text, location, current position, job description, LinkedIn URL, qualification status and the written reason for that status.
Leads Company Details	Company name, website, tagline, description, industry, employee count, locations, specialities, values, target market, key services, tech stack, recent news and the competitor analysis.
Leads Strategy	The generated intelligence document, pain points, solutions and engagement hooks for each qualified lead.
Leads Outreach	Verified email address, the three drafts, the selected email, sent status, sent date, follow up text and follow up status.

Keeping state in Airtable rather than inside the workflow has a second benefit beyond fault tolerance. Every intermediate result is visible to a person. When an email reads badly, the pain points that produced it can be opened and inspected, and the problem can usually be traced to a bad company description three stages earlier.

9. METHODOLOGY

9.1 Map the Manual Process First

The system was not designed by listing available APIs. It was designed by writing out, in order, what a careful researcher would do for one prospect, and then asking which of those steps a machine could do at the same standard. Steps that a machine could do better, such as reading twenty competitor pages, were expanded. Steps that a machine does badly, such as deciding whether a reply is positive, were left out of scope entirely.

This is why the pipeline has a competitor research stage. No manual outbound process at small scale includes competitor research, because it costs too much time. Once the cost drops to a few API calls, it becomes worth doing, and it gives the drafting agent something to say that a template never could.

9.2 Stage Isolation and Flag Based State

Each pipeline reads records where a specific flag is set and its own completion flag is not. Airtable carries fields such as Profile Analysis Complete, Company Analysis Complete, Strategy Generation Complete, Email Drafted, Email Sent and Follow Up Sent. Together these fields act as a state machine held in the database rather than in memory.

The practical effect is that the system is restartable at any point. If the drafting stage is stopped halfway through a run, the records it already handled have Email Drafted set and are skipped on the next run. Nothing is drafted twice and nothing is lost.

9.3 Batching and Rate Control

Ten Split In Batches nodes control how many records are processed per loop iteration. Seven Wait nodes are placed at points where a third party service is being called repeatedly. These pauses are not defensive padding. Apify actor runs and Prospeo lookups both have limits, and the difference between a workflow that finishes and one that is cut off partway is often a two second pause in the right place.

9.4 Handling Unreliable Model Output

Language models return text, and text is not JSON. Four Structured Output Parsers are used to enforce a schema on model output, two of them with auto fix enabled. When the model returns something that does not parse, auto fix sends the output back to the model along with the expected schema and asks for a correction, instead of throwing an error that stops the run. This single setting removed most of the failures seen during testing.

9.5 Duplicate Protection

Duplicates are handled twice, on purpose. A Remove Duplicates node filters within the current batch. Separately, before any lead is created, the workflow searches Airtable for an existing record with the same LinkedIn profile URL and routes to an update rather than a create if one is found. The first check catches repeats inside a run. The second catches repeats across runs, which is the more common case when the same profile appears under two different ICPs.

10. IMPLEMENTATION AND RESULTS

10.1 Pipeline 1: Lead Sourcing and Qualification

This is the largest pipeline at 46 nodes. It starts from a webhook, which allows it to be fired from outside the system, and reads the ICP table for profiles marked as active. A Switch node called Leads Method then routes each profile to one of three sourcing branches based on a Scrape Type field on the record.

The Apify branch calls the LinkedIn Profile Search actor with the job title and filters from the ICP, then passes the resulting profile URLs to the LinkedIn Profile Scraper for full detail. The Google Dork branch takes a different route. An LLM chain generates a set of search operators from the ICP, those operators are run through the Google search actor, and the LinkedIn URLs that come back are scraped. The Prospeo branch calls the search-person endpoint directly with structured filters for job title, company headcount, industry and location, restricted to records that carry a verified email.

Each branch has its own qualification agent, because the shape of the data coming out of the three sources is different. All three agents produce the same structured output: a qualification status of Qualified Lead or Disqualified Lead, and a written reason referring to specific parts of the profile. The agents are instructed to judge on the person's current role rather than their headline, which is a real problem in practice, since LinkedIn headlines are often years out of date.

10.2 Pipeline 2: Company and Competitor Analysis

This pipeline runs every three hours and contains 32 nodes. It pulls qualified leads whose company has not yet been analysed, calls the Apify LinkedIn Company Details actor, and then hands the result to a Company Analysis agent. That agent has two tools available: the Jina Reader endpoint for fetching the company website as clean text, and the Jina Search endpoint for finding additional pages. It produces a structured company record covering description, industry, size, locations, specialities, values, target market, key services and tech stack.

A second agent then handles competitors. It generates search keywords from the company profile, uses Jina Search to find competing businesses, and writes a competitive summary along with improvement recommendations. Four Wait nodes are placed through this pipeline because it makes the heaviest external call volume of any stage.

10.3 Pipeline 3: Profile Intelligence

This pipeline runs hourly across 18 nodes and is where the raw research becomes something usable. A chain called Big Data Analysis takes the full lead record, the company analysis and the competitor analysis, and writes a single markdown intelligence document about the prospect. A second chain reads that document and produces pain points along with matched solutions. A third produces engagement hooks, which are the specific angles the email can open with.

Splitting this into three chains rather than asking for everything at once produced noticeably better output. A single prompt asking for analysis, pain points and hooks together tends to produce three shallow lists. Asking for the analysis first and then reasoning over it gives the model something concrete to work from.

10.4 Pipeline 4: Email Enrichment

Thirteen nodes running every six hours. An agent extracts the clean company domain from the stored website URL, which is less trivial than it sounds because the stored values include tracking parameters, subdomains and occasional LinkedIn redirect links. The cleaned name and domain go to the Prospeo enrich-person endpoint with the only_verified_email flag set. Leads with no verified address are marked and left alone rather than guessed at, since sending to a guessed address damages domain reputation.

10.5 Pipeline 5: Email Drafting

Eleven nodes, running every three hours and also available as a manual run for testing. The drafting agent receives the whole package: name, position, company, headline, background, the intelligence document, the pain points, the solutions and the engagement hooks. It returns three complete drafts that take different angles rather than three rewordings of the same email. All three are stored.

This pipeline is the only one with an explicit error path. A Check If Drafts Generated condition routes failures to a Log Error node, because a silent failure here means a lead sits fully researched and never contacted.

10.6 Pipeline 6: Email Sending

Seven nodes, running once a day at 08:00. An agent reads all three drafts for a lead and selects the one best matched to that prospect, then sends it through a Gmail tool node. The sent email text, the sent status and the sent date are written back to the Leads Outreach table. Sending once a day rather than continuously is a deliberate choice, since a steady low volume looks like a person and a burst looks like a machine.

10.7 Pipeline 7: Follow Up

Nine nodes, running once a day at 10:00. A JavaScript node reads the sent dates, parses them from day/month/year format, and keeps only the leads where three or more days have passed. For each of those, a Gmail search runs against the prospect's address. If any message is found from that address, the lead is skipped.

Only where there is genuine silence does the follow up agent run. It receives all three original drafts and the email that was actually sent, so that the follow up can build on the angle already used rather than starting a fresh pitch. The follow up is sent and the Follow Up Sent flag is written back so it never fires twice.

10.8 Results

Measure	Outcome
Manual time per prospect	Reduced from roughly 15 to 20 minutes of research and writing to zero once an ICP is configured
Sourcing channels	Three, switchable per ICP without any change to the workflow
Research depth per lead	Company profile, competitor analysis, intelligence document, pain points, solutions and engagement hooks
Drafts per lead	Three distinct angles generated, one selected automatically
Duplicate leads created	None observed, due to the two layer duplicate check
Follow ups sent to leads who had replied	None, due to the Gmail reply check before drafting
Recovery after a failed stage	Automatic on the next scheduled run, with no manual restart and no reprocessing of completed records

The result that mattered most during review was not speed. It was that the emails produced referred to things a template could not know: a specific service the company had launched, a gap against a named competitor, a claim on their own website. That is the difference between automated outreach and mass mail.

11. LEARNING OUTCOMES

1. Designing for failure rather than for the happy path. The first version of the pipeline was one long chain and it worked in testing and broke constantly in real runs. Splitting it into seven independent stages with state held in the database was the single change that made the system usable.

2. State belongs in a database, not in a workflow. Once every stage reads and writes flags on a record, the system becomes restartable, inspectable and safe to modify one stage at a time.
3. Prompt decomposition beats prompt length. Three focused prompts chained together produced better output than one long prompt asking for everything, and the intermediate results were also easier to debug.
4. Structured output parsers with auto fix are not optional when a language model sits in the middle of an automated pipeline. Without them, one badly formatted response stops an entire scheduled run.
5. Rate limits shape architecture. Batch sizes and wait placement are not tuning details added at the end, they determine whether a pipeline can finish at all.
6. Data cleaning is most of the work. Extracting a usable company domain from stored website fields took longer to get right than building the agent that used it.
7. Verified data is worth losing volume for. Skipping leads without a verified email address reduces output and protects the sending domain, and that trade is worth making.
8. Working with API documentation, authentication and error responses across five different third party services, and reading a provider's actual behaviour rather than trusting its documentation.

12. CHALLENGES AND SOLUTIONS

12.1 Scraping Without a Session Cookie

Challenge: LinkedIn scraping normally requires a logged in session cookie. A cookie expires, it ties the whole system to one account, and storing it inside a workflow is a security problem. Solution: the system uses cookie free Apify actors from the harvestapi collection for profile search, profile detail and company details. No session token is stored anywhere in the workflow.

12.2 A Single Timeout Wasting an Entire Run

Challenge: the first version was one long workflow. A lead passed through scraping, qualification, research, enrichment, analysis, drafting and sending in a single chain, so one API timeout discarded all the work done before it. Solution: the workflow was split into seven independent pipelines, each with its own trigger, with progress tracked through completion flags in Airtable. A failure now costs one stage rather than the whole run.

12.3 Language Models Returning Invalid JSON

Challenge: language models return text, and text is not always valid JSON. A single malformed response stopped an entire scheduled run. Solution: four Structured Output Parsers enforce a JSON schema on model output, two of them with auto fix enabled. Malformed output is returned to the model with the expected schema attached and corrected, instead of raising an error.

12.4 Duplicate Lead Records

Challenge: the same LinkedIn profile often appears under more than one customer profile, which created duplicate records. Solution: two layers of protection were added. A Remove Duplicates node filters within the current batch, and before any lead is created the workflow searches Airtable by profile URL and routes to an update when a match is found.

12.5 Third Party Rate Limits

Challenge: both Apify and Prospeo apply rate limits, and batch runs were being cut off partway through. Solution: batch sizes were reduced through Split In Batches nodes and seven Wait nodes were placed at the points where external services are called most heavily.

12.6 Unusable Company Website Values

Challenge: stored company website fields contained tracking parameters, subdomains and occasional LinkedIn redirect links, so email enrichment lookups failed. Solution: an agent step was added that extracts and normalises the domain before the enrichment call, with a structured output schema so the result is always a clean domain string.

12.7 Follow Ups Sent to Prospects Who Had Replied

Challenge: a scheduled follow up risked being sent to someone who had already answered the first email. Solution: a Gmail search runs against the prospect's address before any follow up is drafted. If any message is found from that address, the lead is skipped entirely.

12.8 Date Parsing and Shallow Prompt Output

Challenge: sent dates were stored as day/month/year strings, which JavaScript does not parse correctly by default, and a single prompt asking for analysis, pain points and hooks together produced shallow, generic output. Solution: a code node was written that splits the date string and constructs the date explicitly, normalising both dates to midnight before comparing. The prompt was split into three chained stages, each reading the output of the previous one, so that later stages reason over concrete material instead of raw profile data.

13. CONCLUSION

The twelve week internship at CashCow Labs produced a working Lead Generation and Outreach System built on n8n, with 179 nodes across seven independent pipelines, integrating Apify, Prospeo, Jina AI, OpenAI, Airtable and Gmail. The system takes a customer profile as input and produces researched, personalised, sent and followed up cold emails as output, with no manual step in between.

The technical learning was real, but the more useful lesson was about design. The first version of this system was a single long workflow that did everything in order. It demonstrated well and failed in production, because it assumed every external service would respond and every model would return valid JSON. The version described in this report assumes the opposite, and that is why it runs.

Working on a system that other people depend on is different from working on an assignment. An assignment is finished when it is submitted. This system is finished when it can run for a week without anyone looking at it, and getting to that point took more time than building the features did.

14. FUTURE SCOPE

1. Reply classification. The system currently detects that a reply exists and stops. A classification stage could separate interested, not interested and out of office replies, and route each differently.
2. Multi channel sequencing. Adding LinkedIn connection requests and messages alongside email, with the channels aware of each other so a prospect is not contacted twice on the same day.
3. A feedback loop on draft selection. The system stores three drafts and picks one. Recording which selections led to replies would let the selection step learn from outcomes rather than judging each email in isolation.
4. Deliverability handling inside the workflow, including bounce detection, sending domain rotation and volume ramping.
5. A reporting layer over the Airtable base showing reply rate by ICP, by sourcing method and by email angle, so that the profile definitions can be tuned on evidence.
6. Cost tracking per lead across Apify, Prospeo, Jina and OpenAI, which would make it possible to compare the three sourcing routes on cost per qualified lead rather than on volume.
7. Human review checkpoints as an optional mode, where drafts wait for approval in Airtable before sending, for clients who want oversight on their first campaigns.

15. BIBLIOGRAPHY AND REFERENCES

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16. ANNEXURE

- Internship Completion Certificate, attached at the front of this report.
- Exported workflow file, leadgen_workflow.json, containing all 179 nodes.
- Airtable base structure and field listing, given in Annexure A.
- External services and endpoints used by the system, given in Annexure B.
- Project repository containing the report, the presentation, the certificate and the workflow export.

Annexure A: Airtable Base Structure

Base name: Lead Generation and Outreach System. Five linked tables, with the key fields in each listed below.

Table	**Key Fields**
ICP	ICP Name, Query, Job Title, Decision Maker, Location, Company Size, Industry, Scrape Type, Google Dork Query, Angles, Generate Leads for this ICP?
Lead Qualification	Full Name, Headline, About, Location, Company Name, Current Position, Current Job Description, Linkedin Profile Url, Qualification status, Reason of Disqualification/Qualification
Leads Company Details	Company Name, Company Website Link, Company Tagline, Company Description, Company Industry, Employee Count, Company Locations, Company Specialities, Company Values, Target Market, Key Services, Tech Stack, Recent News, Competitor Analysis, Competitive Summary, Improvement Recommendations, Company Analysis Complete, Competitor Analyzed
Leads Strategy	Full Profile + Company + Competitor Analysis, Pain Points, Solutions, Strategy Hooks, Profile Analysis Complete, Strategy Generation Complete
Leads Outreach	Full Name, Email ID, Draft 1, Draft 2, Draft 3, Qualified Email, Email Drafted, Email Sent, Email Sent Date, Email Status, Follow Up Email, Follow Up Sent

Annexure B: External Services and Endpoints

Service	Endpoint or Actor	Used For
Apify	harvestapi/linkedin-profile-search	Finding profiles that match a job title and filter set
Apify	harvestapi/linkedin-profile-scraper	Full profile detail without a session cookie
Apify	harvestapi/linkedin-company	Company page detail in bulk
Apify	tuningsearch/cheap-google-search-results-scraper	Running generated search operators
Prospeo	POST /search-person	People search by title, headcount, industry and location
Prospeo	POST /enrich-person	Verified email lookup from name and company domain
Jina AI	POST s.jina.ai	Search, used as an agent tool
Jina AI	POST r.jina.ai	Page fetch returning clean text, used as an agent tool
OpenAI	gpt-4.1-mini, gpt-4o-mini	Qualification, research, analysis, drafting and follow up

Annexure C: Project Repository

The complete workflow export, this report, the presentation and the internship certificate are published in the project repository. The exported JSON in the repository has all API keys and credentials removed and replaced with placeholders.

Annexure D: Internship Certificate

A copy of the internship completion certificate issued by CashCow Labs is included at the front of this report and in the project repository.